

Deriving Causal Interpretations from fMRI during Binocular Rivalry

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Motivation

Context

- fMRI data only reveals correlations between stimuli and brain activity; due to potential confounders, these associations may be spurious
- Causal structure of the underlying process remains unclear

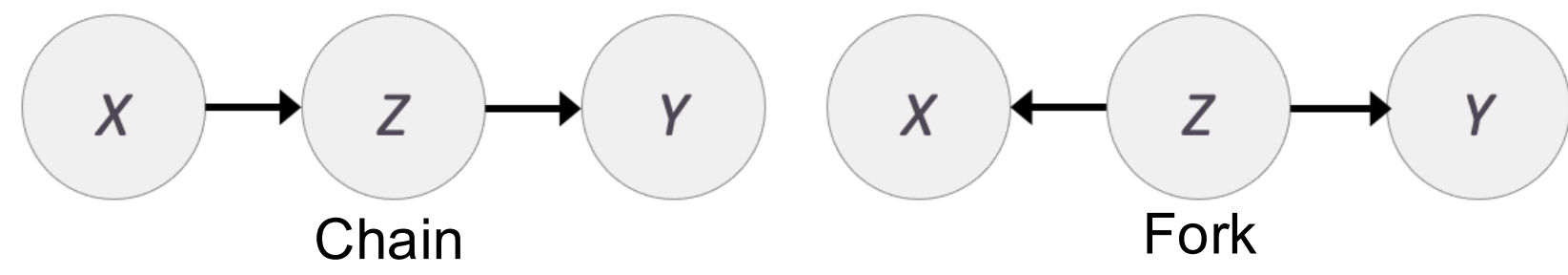
Approach

- Causal Bayesian Networks (CBNs) in combination with encoding and decoding model allow systematic testing of causal relations
- Thereby, directionality and vicinity of relations can be determined

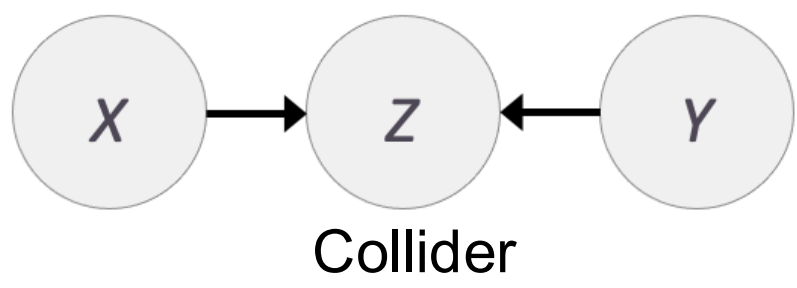
Background

Directed Acyclic Graphs (DAGs)

Conditional Independence:

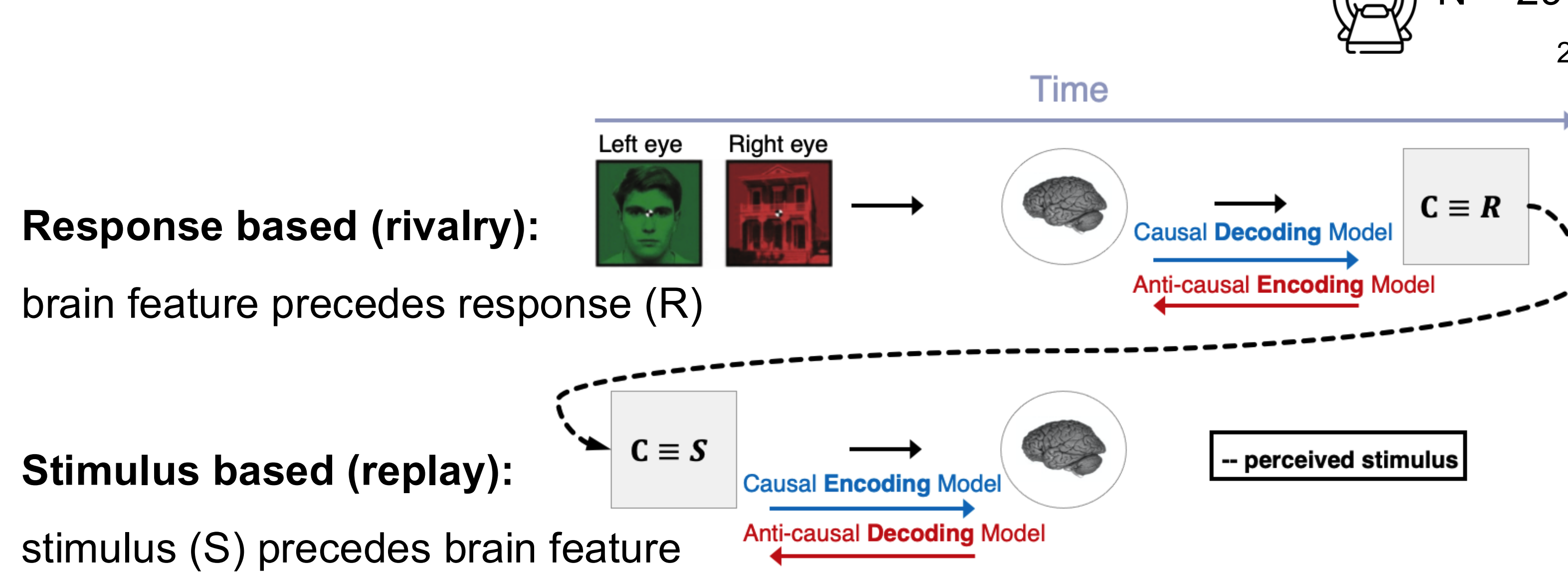


Conditional Dependence:

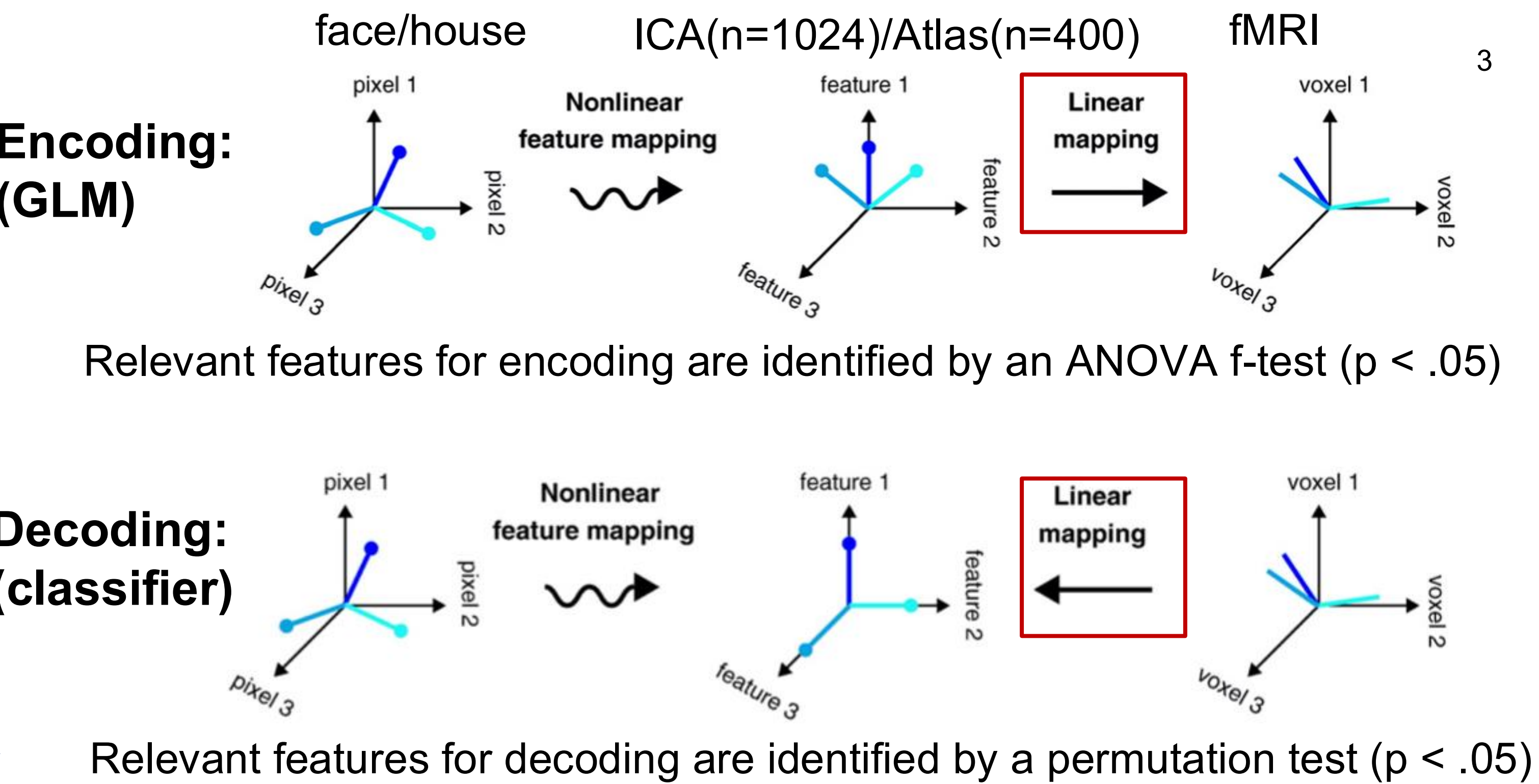


- Dependency separation: D-separation is when conditioning on a node or set of nodes results in two other nodes becoming independent
- Causal Markov Condition: The CMC expresses the notion that each node in a causal DAG becomes independent of its non-descendants given its direct causes
- Faithfulness: The faithfulness assumption states that all (conditional) independences between the random variables of a DAG are implied by its causal structure

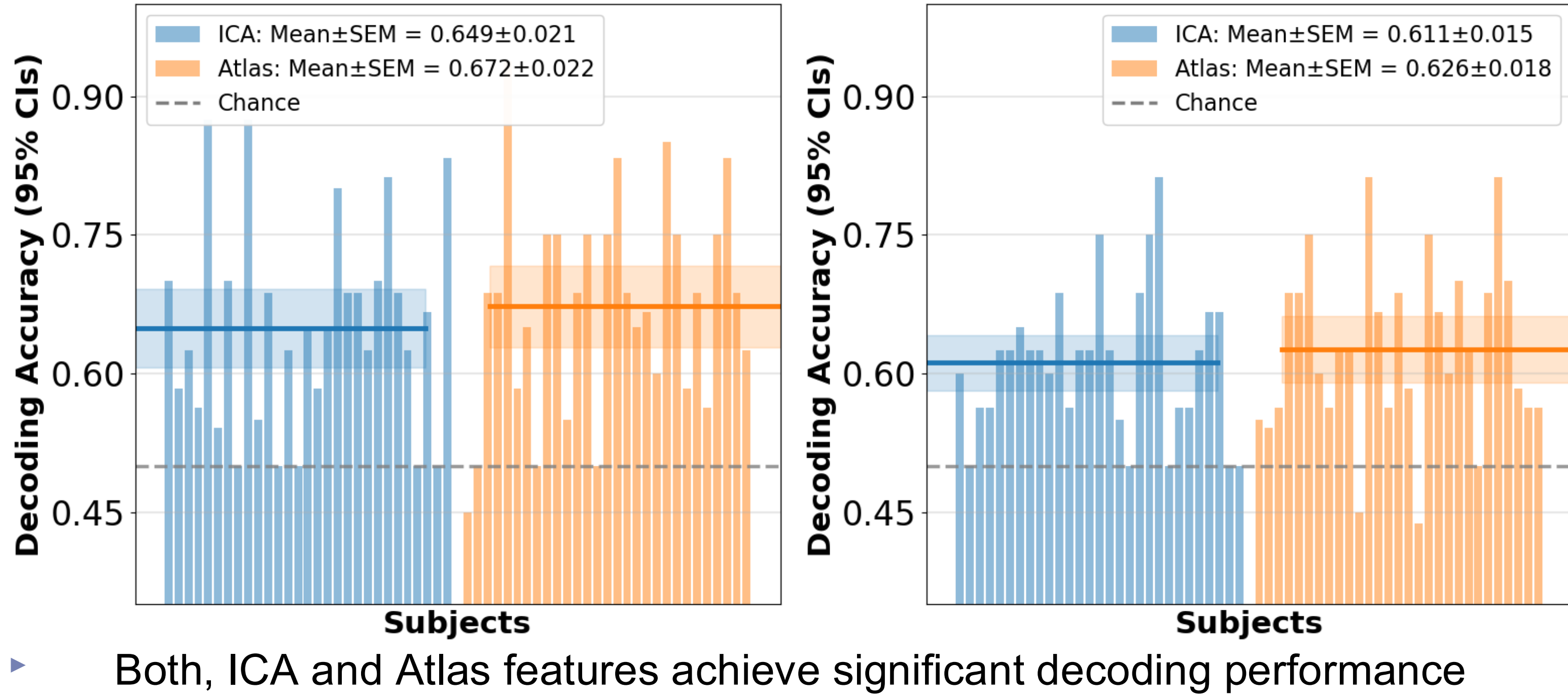
Experimental Paradigm



1. Is information about stimuli captured in features of encoding and decoding models?



Group Level Decoding Performance of Face vs House Stimulus

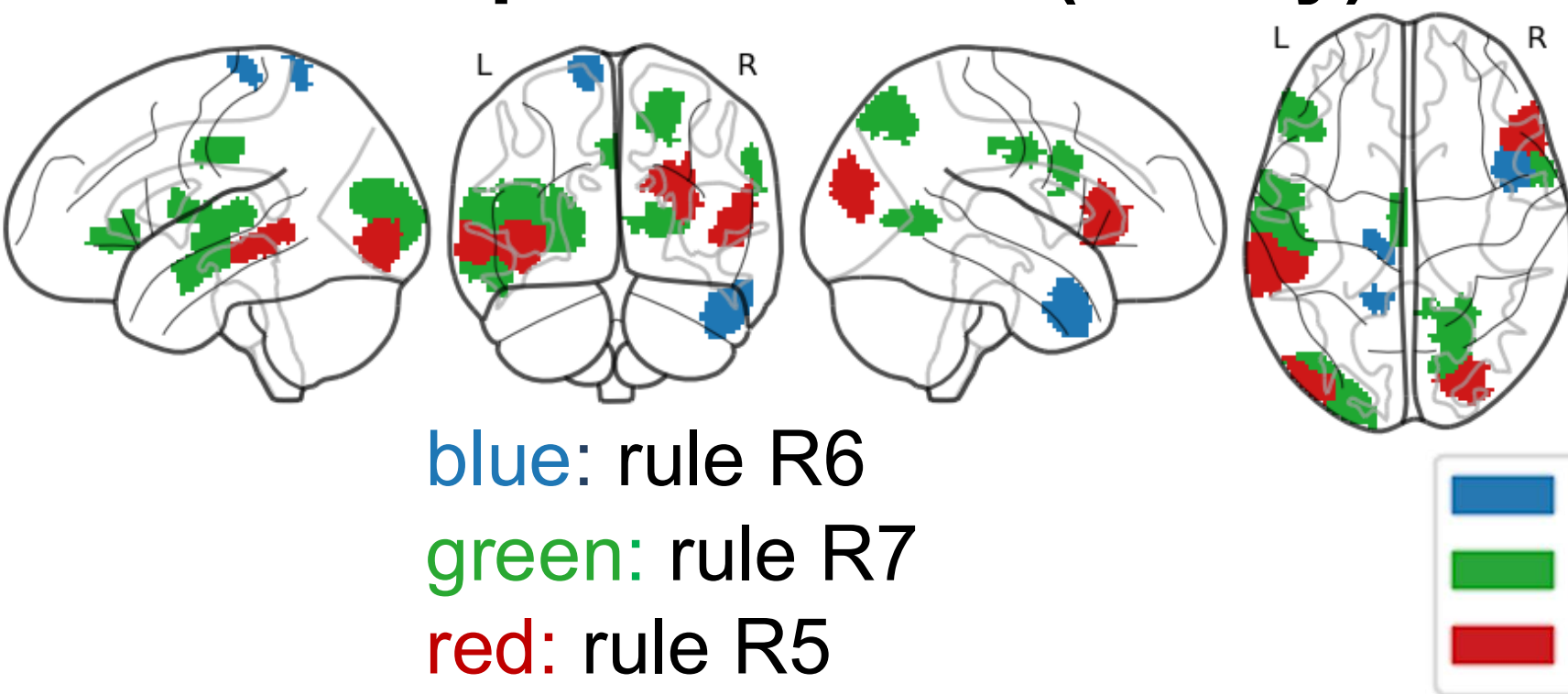


2. Are causal relations present on the individual level?

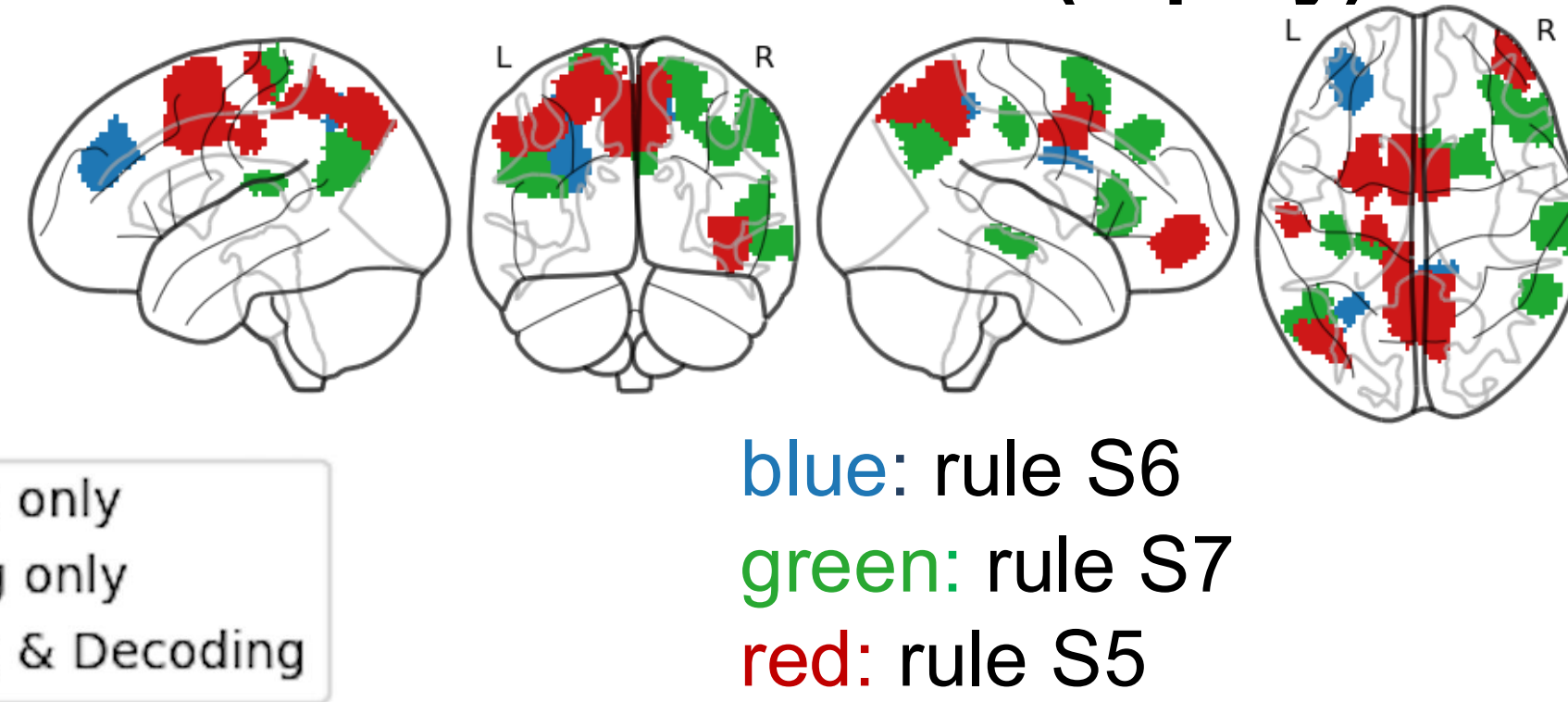
Interpretations Rules for Brain Features

	encoding	decoding	causal interpretation	rule
stimulus-based	✓	n/a	X_i effect of s	S1
	×	n/a	X_i no effect of s	S2
	n/a	✓	inconclusive	S3
	n/a	×	inconclusive	S4
	✓	✓	X_i effect of s	S5
	✓	×	X_i indirect effect of s	S6
	×	✓	provides brain state context	S7
	×	×	neither effect nor provides brain state context	S8
response-based	✓	n/a	inconclusive	R1
	×	n/a	X_i no cause of r	R2
	n/a	✓	inconclusive	R3
	n/a	×	inconclusive	R4
	✓	✓	inconclusive	R5
	✓	×	X_i no direct cause of r	R6
	×	✓	provides brain state context	R7
	×	×	neither effect nor provides brain state context	R8

Response based (rivalry)



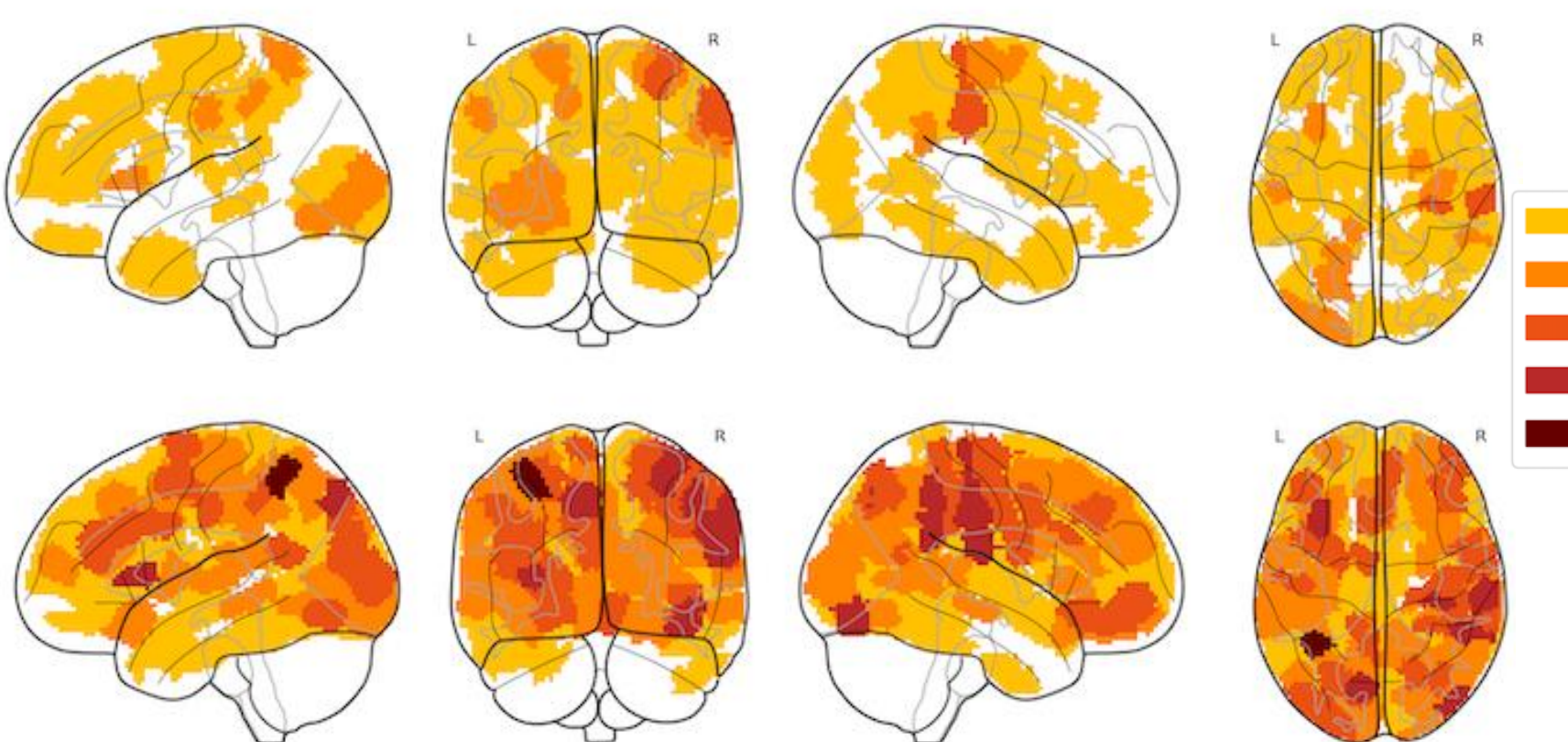
Stimulus based (replay)



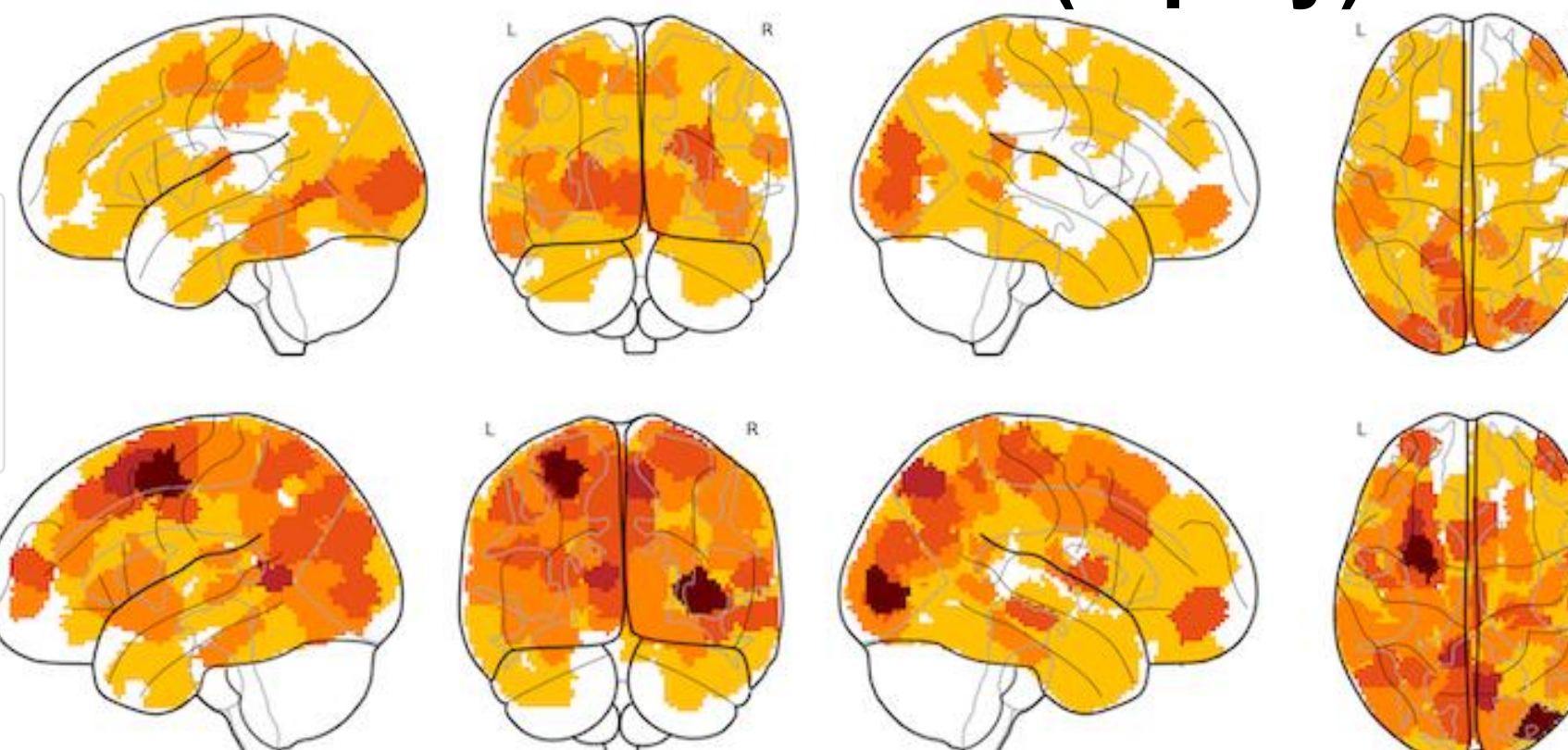
- For this subject sig. features are marked, for some subjects no sig. features are found

3. Do causal relations generalize on the group level?

Response based (rivalry)



Stimulus based (replay)



- Encoding (upper row) and decoding (lower row) exhibit strong interindividual variance

Conclusions

- Applying encoding and decoding models to fMRI data allows to infer causal relations between stimuli and brain features
- Interindividual differences are very pronounced, for most subjects we find relevant features for encoding and decoding but not for all subjects
- Relevant features in the DIPFC have been identified for a subset of subjects, indicating stimulus awareness and conscious processing
- Because of the strong interindividual differences, we do not find clear effects on the group-level

References

¹ Mumford, J. A., & Ramsey, J. D. (2014). Bayesian networks for fMRI: a primer. *Neuroimage*, 86, 573-582.
² Schölkopf, B., Janzing, D., Peters, J., Sgouritsa, E., Zhang, K., & Mooij, J. (2012). On causal and anticausal learning. *arXiv preprint arXiv:1206.6471*.
³ Naselaris, T., Kay, K. N., Nishimoto, S., & Gallant, J. L. (2011). Encoding and decoding in fMRI. *Neuroimage*, 56(2), 400-410.
⁴ Weichwald, S., Meyer, T., Özdenizci, O., Schölkopf, B., Ball, T., & Grosse-Wentrup, M. (2015). Causal interpretation rules for encoding and decoding models in neuroimaging. *Neuroimage*, 110, 48-59.